ICL-Traffic Forecasting: Lightweight In-Context Learning for Pretrained Spatio-Temporal Foundation Models

Parth Ripal Parikh
North Carolina State University
Raleigh, NC, USA
pparikh2@ncsu.edu

Mrudani Vishal Hada
North Carolina State University
Raleigh, NC, USA
mhada2@ncsu.edu

Xuesi Zhou
North Carolina State University
Raleigh, NC, USA
xzhou38@ncsu.edu

Ranga Raju Vastavai
North Carolina State University
Raleigh, NC, USA
rrvastav@ncsu.edu

Abstract

Pretrained spatio-temporal foundation models (ST-FMs) such as OpenCity deliver strong zero-shot traffic forecasts, yet accuracy drops under cross-city distribution shift. In-context learning (ICL) adapts predictions at inference time from retrieved demonstrations but has not been studied for ST-FMs. The standard approach of feeding demonstrations into the model context fails here: these models are pretrained on fixed-length sequences, and extending the input degrades MAE by 9–28%. We propose **ICL-Traffic**, a lightweight output-space ICL module that keeps the base model completely frozen. A 198K-parameter cross-attention aggregator trains offline in seconds from cached features; at inference time, KNN retrieves similar training windows and the aggregator combines their prediction errors into a per-query correction with no test-time gradients. On PEMS07M, ICL-Traffic reduces MAE from 4.50 to 4.15 (7.8%) with only ∼0.8% extra parameters. On two out-of-distribution datasets (SZ-DIDI and CD-DIDI), the module recovers 78–84% of the MAE gain from prediction-head fine-tuning, without modifying a single backbone weight.

CCS Concepts

• **Computing methodologies** → **Machine learning**.

Keywords

traffic forecasting, in-context learning, foundation models, spatio-temporal prediction, retrieval-augmented inference

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1 Introduction

Accurate short-term traffic forecasting underpins intelligent transportation systems, from adaptive signal control to congestion management. Classical models such as STGCN [13] and Graph WaveNet [12] perform well but require per-dataset training. Spatio-temporal foundation models (ST-FMs) address this by pretraining on large, multi-city corpora. OpenCity [8], which combines a Transformer backbone with graph neural networks, achieves strong zero-shot performance across cities and data types, but accuracy degrades under cross-city shift: on the ride-hailing datasets SZ-DIDI and CD-DIDI, its zero-shot MAE of 4.66 and 6.34 trails supervised baselines substantially. Closing this gap with 3-epoch prediction-head fine-tuning requires gradient updates, raising a natural question: can demonstration-based adaptation at inference time approximate fine-tuning's gains while keeping the backbone entirely frozen?

In NLP, ICL lets foundation models adapt via demonstrations without weight updates [1, 2]. Das et al. [3] extend this idea to time series, adapting TimesFM [4] through cross-example attention. Both approaches inject demonstrations directly into the input context, an assumption requiring 1-D causal sequences and architectural retraining. Neither assumption holds for ST-FMs, which operate on structured spatio-temporal graphs and are pretrained on fixed-length sequences (24 patches for OpenCity). We confirm empirically that naively extending the input context distorts attention patterns and degrades MAE by 9–28%. Output-space ICL, where demonstrations never enter the context and corrections happen after prediction, remains unexplored for ST-FMs.

We propose **ICL-Traffic**, a residual correction framework for any frozen ST-FM. The core insight is that a demonstration's prediction error estimates the frozen model's bias on similar inputs, and a learned aggregation network can combine these errors across K retrieved demonstrations into a query-specific correction. The base model stays frozen, and the aggregator trains offline in seconds from cached features. Our contributions are: (i) the first output-space, gradient-free ICL framework for frozen ST-FMs, requiring no backbone weight changes and enabling lightweight deployment, in contrast to FlashST [9] and UniST [14], which update parameters at adaptation time; (ii) a systematic analysis showing that retrieval and learned aggregation are jointly necessary, where retrieval supplies the error signal and the aggregator filters noise across demonstrations of varying relevance; and (iii) empirical gains on PEMS07M

(7.8% MAE reduction, 198K parameters) and OOD cities (recovering ≈80% of prediction-head fine-tuning gains), together with the empirical gap, which quantifies the boundary where output-space corrections must yield to representation-level alignment.

2 Related Work

2.1 Spatio-Temporal Foundation Models

Classical spatio-temporal models, including STGCN [13], Graph WaveNet [12], and ASTGCN [6], learn per-dataset parameters and do not transfer across cities. OpenCity [8], our base model, combines a Transformer with graph neural networks, using instance normalization and temporal-categorical attention to reach zero-shot accuracy competitive with supervised baselines across multiple benchmarks. UniST [14] uses knowledge-guided prompts across urban scenarios, FlashST [9] trains a spatio-temporal prompt network on downstream data, and UniFlow [15] retrieves structured memory of shared spatio-temporal patterns; all three, however, modify parameters during adaptation. None operate under the frozen-backbone, output-space constraint we target.

2.2 In-Context Learning

ICL, introduced by GPT-3 [1], enables language models to adapt to new tasks from a handful of demonstrations without weight updates. Chen et al. [2] formalize this as In-Context Tuning, meta-training on concatenated demonstrations and queries, which outperforms gradient-based meta-learning [5] while being less sensitive to example ordering. Das et al. [3] extend ICL to time series, adapting TimesFM [4] through cross-example attention. Both approaches inject demonstrations into the model context and require continued pretraining, a strategy that fails for ST-FMs as we show empirically, and one that does not naturally extend to structured spatio-temporal graphs.

2.3 Test-Time Adaptation

Test-time adaptation (TTA) updates model parameters at inference using self-supervised signals: TENT [11] minimizes prediction entropy over batch-norm parameters, and TTT [10] adds an auxiliary self-supervised head. Both require backward passes at test time. ICL-Traffic instead trains its aggregator once offline from cached features, with zero test-time gradients. TTA's unsupervised signals also assume covariate shift, which breaks down when a model trained on highway speeds is applied to ride-hailing demand in a new city. ICL-Traffic uses labeled target-domain demonstrations to estimate this bias directly.

3 Method

3.1 Problem Formulation

A traffic network has N sensor nodes. Given historical observations $\mathbf{X} \in \mathbb{R}^{T\times N\times F}$, the task is to predict future values $\mathbf{Y} \in \mathbb{R}^{T'\times N}$. Let f_θ be a frozen pretrained ST-FM producing $\hat{\mathbf{Y}} = f_\theta(\mathbf{X})$, and let $\mathcal{D} = \{(\mathbf{X}_d, \mathbf{Y}_d)\}_{d=1}^{|\mathcal{D}|}$ be a demonstration pool drawn from the target-domain training split. For a test query $\mathbf{X}_q$, we select K demonstrations from $\mathcal{D}$ and produce a prediction $\hat{\mathbf{Y}}_q$ that improves on $f_\theta(\mathbf{X}_q)$ without updating θ.

3.2 Residual Correction Framework

A demonstration's known prediction error is a direct signal of the frozen model's bias on similar inputs. Aggregating these errors across retrieved demonstrations yields a query-specific correction without any gradient step. For each demonstration $(\mathbf{X}_k, \mathbf{Y}_k)$, we compute the residual

$$\mathbf{c}_k \;=\; \mathbf{Y}_k - f_\theta(\mathbf{X}_k), \tag{1}$$

and correct the query prediction as

$$\hat{\mathbf{Y}}_q = f_\theta(\mathbf{X}_q) + \mathrm{Agg}\Big(\{\mathbf{c}_k\}_{k=1}^{K},\, \mathbf{h}_q,\, \{\mathbf{h}_k\}_{k=1}^{K}\Big), \tag{2}$$

where $\mathbf{h}_q$ and $\mathbf{h}_k$ are frozen-encoder features. Every input passes through the standard forward path, so sequence length is unchanged, and K=0 recovers zero-shot prediction. Demo predictions and encoder features are cached in a one-time Phase 1 pass, so training the aggregator requires no additional forward passes through the base model.

3.3 Demonstration Retrieval

Three retrieval strategies are considered. *Random selection* samples K demonstrations uniformly from $\mathcal{D}$; since their residuals are uncorrelated with the query's error, they contribute noise rather than signal. *Raw-input KNN* retrieves the K nearest neighbors by L2 distance in the flattened input space,

$$\mathcal{N}_K(\mathbf{X}_q) \;=\; \arg\min_{\mathcal{S}\subset\mathcal{D},\,|\mathcal{S}|=K} \sum_{d\in\mathcal{S}} \big\|\mathrm{vec}(\mathbf{X}_q) - \mathrm{vec}(\mathbf{X}_d)\big\|_2, \tag{3}$$

a clear improvement over random selection but one that still treats every element as an independent coordinate. *Embedding-space KNN* retrieves neighbors in the frozen encoder's pooled feature space,

$$\bar{\mathbf{h}} \;=\; \tfrac{1}{N}\sum_{n=1}^{N} \mathbf{h}_n \;\in\; \mathbb{R}^{512}, \tag{4}$$

$$\mathcal{N}_K(\mathbf{X}_q) \;=\; \arg\min_{\mathcal{S}} \sum_{d\in\mathcal{S}} \big\|\bar{\mathbf{h}}_q - \bar{\mathbf{h}}_d\big\|_2, \tag{5}$$

which better captures shared congestion regimes and rush-hour structure, in the spirit of kNN-LM [7]. We use embedding-space KNN as the in-distribution retriever on PEMS07M, but retain raw-L2 KNN for the out-of-distribution cities, where severe domain shift can distort the encoder's representation space enough to select misaligned demonstrations (Section 4.4).

Temporal exclusion protocol. The demonstration pool $\mathcal{D}$ is built exclusively from each dataset's training split, and all evaluations run on disjoint test splits. During Phase-2 aggregator training, we additionally exclude any demonstration whose window temporally overlaps the training query (defined as $|d_{\text{start}} - q_{\text{start}}| < T + T'$). Resulting pool sizes are 5,761 windows for PEMS07M and 8,353 each for SZ-DIDI and CD-DIDI.

3.4 Cross-Attention Aggregator

Naive uniform averaging assigns equal weight $1/K$ to every demonstration's residual, diluting informative corrections with noise, and on its own it never surpasses zero-shot. We compare a minimal 8K-parameter cosine-similarity variant (SimpleDemoAggregator) against our main 198K-parameter cross-attention design (DemoAggregator).

The aggregator takes three cached Phase-1 tensors as input: query features $\mathbf{h}_q$ (Q), demo features $\{\mathbf{h}_k\}$ (K), and demo residuals $\{\mathbf{c}_k\}$ (V). Layer-normalized Q/K projections across H=4 heads produce a per-node softmax attention weight $\alpha_{b,n,h,k}$ over the K demonstrations, because different sensors may match different demonstrations most closely. We skip the standard value projection: the demo residuals already live in prediction space, so the attention weights act on them directly, are merged across heads by a bias-free linear layer, and are scaled by a query-conditioned gate $g_n \geq 0$ (a two-layer MLP with softplus output, zero-initialized so $g_n \approx 0.693$ at training start, chosen over sigmoid to allow corrections to be amplified under severe shift). The final correction and prediction are

$$\Delta_q = g \cdot \tilde{\mathbf{c}}, \qquad \hat{\mathbf{Y}}_q = f_\theta(\mathbf{X}_q) + \Delta_q. \tag{6}$$

The aggregator has ≈198K parameters (≈0.8% of the 26M-parameter OpenCity-plus backbone).

3.5 Training Procedure

Training has two phases. *Phase 1* (one-time) runs the frozen backbone over all training windows and caches, for each $d \in \mathcal{D}$, the prediction $f_\theta(\mathbf{X}_d)$, encoder features $\mathbf{h}_d$, and residual $\mathbf{c}_d$. *Phase 2* trains only the aggregator from this cache using Adam (10^{-4}), MAE loss, and early stopping with patience 5, with no forward or backward pass through the base model. Because Phase 1 is amortized across all downstream configurations, all experiments reuse the same cache.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on three traffic datasets. *PEMS07M* contains 228 highway speed sensors in California at 5-minute intervals (50/10/40 train/val/test split, 5,761 training windows); it is absent from OpenCity's pretraining corpus but shares the highway-speed modality, making it our in-distribution benchmark. *SZ-DIDI* (Shenzhen, 627 nodes) and *CD-DIDI* (Chengdu, 524 nodes) are ride-hailing demand datasets at 10-minute intervals, representing an out-of-distribution regime with unseen cities, an unseen traffic modality, and a coarser temporal resolution.

Base model. We use OpenCity-plus [8] with embedding dimension 512, encoder depth 6, 16 attention heads, and ∼26M parameters. For DIDI datasets, input and output windows are set to 144 steps with `PatchEmbedding` gap = 2, following the OpenCity recipe.

Metrics and implementation. We report MAE, RMSE, MAPE (%), and Pearson correlation (CORR), using standard masked evaluation (mae threshold = 0, mape threshold = 0.001). Experiments use PyTorch 2.4.1 on an NVIDIA L40S (46 GB). Datasets and code are publicly available at https://huggingface.co/datasets/hkuds/OpenCity-dataset/tree/main and https://github.com/Parth291102/Traffic-Forecasting-TSFM.git.

4.2 Main Results on PEMS07M

Table 1 summarizes our PEMS07M results. Random demonstrations substantially degrade performance (MAE 4.50 → 6.14), confirming that uncorrelated residuals introduce noise. Raw-input KNN reduces this degradation (MAE 6.14 → 4.66) but does not surpass zero-shot, showing that retrieval alone is insufficient. Learned aggregation pushes beyond zero-shot: SimpleDemoAggregator and DemoAggregator achieve MAE 4.47 and 4.30, respectively. Using embedding-space KNN with the cross-attention aggregator yields the best result (MAE 4.15), a 7.8% improvement over zero-shot with only 0.8% additional parameters and no backbone modification.

Table 1: Main results on PEMS07M. All ICL methods use K=3. Best in bold. Dashes for full-shot baselines indicate metrics not reported in [8].

Method	Retrieval	Params	MAE↓	RMSE↓	MAPE%↓	CORR↑
Zero-Shot	—	0	4.50	8.21	12.20	0.736
ICL-Traffic / Random	random	0	6.14	9.53	14.84	0.634
ICL-Traffic / Naïve avg	raw-L2	0	4.66	8.11	11.92	0.729
ICL-Traffic / Simple agg	raw-L2	8K	4.47	8.01	11.55	0.737
ICL-Traffic / Cross-attn (3ep)	raw-L2	198K	4.30	7.68	11.37	0.748
ICL-Traffic / Cross-attn (5ep)	**emb-KNN**	198K	**4.15**	**7.61**	**11.25**	**0.739**
Full-shot supervised baselines (as reported by [8])						
GWN [12]	—	full	4.17	—	—	—
ASTGCN [6]	—	full	4.39	—	—	—
STGCN [13]	—	full	4.44	—	—	—

Table 2: Cross-dataset adaptation. ICL aggregator and fine-tune both use 3 epochs on target-domain training data. Best in bold, best frozen-backbone in underline.

Dataset	Method	Backbone	MAE↓	RMSE↓	MAPE%↓	CORR↑
SZ-DIDI	Zero-Shot	frozen	4.66	7.10	19.05	0.322
	ICL-Traffic	frozen	2.70	4.11	11.08	0.540
	Fine-tune head (3ep)	modified	**2.33**	**3.75**	**9.89**	**0.681**
CD-DIDI	Zero-Shot	frozen	6.34	9.36	28.36	0.336
	ICL-Traffic	frozen	3.41	5.02	15.28	0.720
	Fine-tune head (3ep)	modified	**2.60**	**3.93**	**11.98**	**0.793**

4.3 Cross-Dataset Generalization: ICL vs. Fine-Tuning

Table 2 and Fig. 1 compare ICL-Traffic against gradient-based prediction head fine-tuning on SZ-DIDI and CD-DIDI. Both methods use the same target-domain training data and 3-epoch adaptation budget; the backbone remains frozen for ICL-Traffic and only the prediction head is modified for fine-tuning.

ICL-Traffic's relative improvement over zero-shot scales with distribution shift. On PEMS07M, the aggregator reduces MAE by 4.4–7.8%, whereas on SZ-DIDI and CD-DIDI the gains reach 42% and 46%, respectively. This indicates that KNN demonstrations expose systematic prediction biases under distribution shift, enabling effective output-space correction without modifying the backbone. We quantify the fraction of fine-tuning's improvement recovered by ICL as

$$\text{Recovery} = \frac{\text{MAE}_{\text{ZS}} - \text{MAE}_{\text{ICL}}}{\text{MAE}_{\text{ZS}} - \text{MAE}_{\text{FT}}}. \tag{7}$$

ICL-Traffic recovers 84% of the fine-tuning gain on SZ-DIDI and 78% on CD-DIDI while keeping the backbone frozen.

4.4 Analysis

Random demonstrations provide corrections largely uncorrelated with the query's prediction error, so averaging them introduces

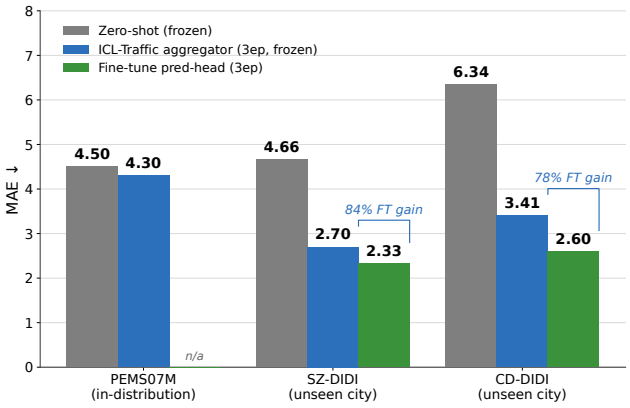


Figure 1: Cross-dataset adaptation on SZ-DIDI and CD-DIDI (3-epoch budget): zero-shot, frozen-backbone ICL-Traffic, and prediction-head fine-tuning.

noise rather than signal. KNN retrieval identifies demonstrations with correlated errors, providing a useful residual signal, but uniform averaging dilutes this signal when retrieved demonstrations vary in relevance; learned aggregation resolves this by assigning query- and node-dependent weights. Across all K values tested, performance improves sharply from K=1 to K=3 and then plateaus: MAE changes from 7.39 at K=1 to 4.30 at K=3, with K=5 and K=7 achieving 4.28 and 4.29, respectively (all with cross-attention and raw-L2 retrieval). We therefore use K=3 throughout. We also evaluated standard sequence-level ICL by concatenating demonstrations with the model input, which consistently degraded MAE by 9–28% relative to zero-shot due to distributional and positional-encoding mismatches, confirming that output-space correction is essential for frozen ST-FMs.

4.5 Limitations

ICL-Traffic is more precisely a retrieval-augmented residual-correction method than input-space ICL in the sense of ICT [2] or TimesFM-ICF [3], though it shares the gradient-free, demonstration-based adaptation goal that motivates the ICL framing. The remaining ≈20% gap relative to fine-tuning is an empirical observation specific to this method and baseline; retrieval quality, aggregator capacity, or a stronger fine-tuning baseline could each narrow it. Retrieval and aggregation also treat sensor networks as flat node collections without exploiting graph topology, and reported numbers reflect single-seed runs without variance quantification.

4.6 Conclusions

ICL-Traffic demonstrates that frozen ST-FMs can be meaningfully adapted through output-space, demonstration-based correction without gradient updates, architectural changes, or backbone retraining. KNN retrieval and a lightweight cross-attention aggregator together constitute an effective surrogate for gradient-based adaptation, recovering up to 84% of fine-tuning's MAE gain with only 0.8% additional parameters and zero test-time gradients. The benefits scale with distribution shift severity, suggesting that systematic prediction biases under domain shift are particularly amenable to output-space correction. The remaining ≈20% gap motivates future work combining ICL-Traffic with lightweight representation-level adaptation and graph-aware retrieval over the sensor adjacency structure. We release our code and model weights to support reproducibility and extension to other ST-FM architectures.

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